

Multi-Objective Reinforcement Learning for RNA Inverse Folding: Balancing Thermodynamic Stability and Sequence Viability

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Abstract

The RNA inverse folding problem is a foundational combinatorial optimization challenge in computational biology, which demands the discovery of a discrete nucleotide sequence that autonomously folds into a target secondary structure. Although existing Deep Reinforcement Learning (DRL) frameworks, such as LEARNA, have successfully treated this as a single-objective thermodynamic optimization problem, they neglect important sequence-level properties that determine whether a designed molecule is synthesizable and biophysically stable. Sequences optimized solely for structural match may exhibit extreme GC-content, long homopolymer runs that impede chemical synthesis, or weak thermodynamic stability despite correct topology. In this work, we propose a multi-objective DRL approach extending the LEARNA environment. We train agents to navigate the combinatorial search space 4^n to design RNA sequences that not only meet structural targets but also maintain a biologically plausible GC-content (40–60%), minimize homopolymer runs, and maximize thermodynamic stability via minimum free energy (MFE). We employ an adaptive reward weighting strategy to handle the inherent tension between these objectives, and systematically compare Value-based (DQN) and Policy Gradient (PPO) methods on sample efficiency, convergence stability, and the resulting Pareto trade-offs.

1 Introduction and Motivation

The design of functional RNA molecules is central to modern biotechnology, especially in the development of mRNA vaccines, ribozymes, and targeted therapeutics. Because the biological function of an RNA molecule is fundamentally determined by its secondary and tertiary structure, synthetic biology frequently encounters the **RNA Inverse Folding Problem**: given a desired, functional 2D structural shape, what is the exact sequence of discrete nucleotides (A, C, G, U) that will thermodynamically fold into that specific shape?

From a computational perspective, this is an NP-hard Combinatorial Optimization problem. For a target structure of length n , the algorithm must select from an alphabet of 4 nucleotides at every position, creating a search space of 4^n . A target of merely 50 nucleotides presents over 1.2×10^{30} possible combinations. Traditional heuristic algorithms often become trapped in local thermodynamic optima within this super-exponential space, a phenomenon well-documented in the Eterna massive open laboratory studies [Anderson-Lee et al., 2016].

Recently, Runge et al. [2019] demonstrated that Deep Reinforcement Learning (DRL) can effectively solve this combinatorial problem by treating nucleotide selection as a sequential decision-making process. Their open-source LEARNA framework trains agents to design sequences that minimize the thermodynamic free energy distance to the target structure. However, structural match alone does not guarantee a useful molecule. A sequence that folds correctly may still be impractical if it has extreme GC-content (making it prone to misfolding or aggregation), contains long homopolymer runs (which hinder chemical synthesis and sequencing accuracy [Leppik et al., 2022]), or achieves the correct topology with a shallow free-energy minimum (making it thermodynamically fragile).

These sequence-level quality metrics, GC-content, homopolymer avoidance, and MFE magnitude, are universally applicable to *all* RNA classes, whether coding or non-coding. Unlike protein-specific metrics such as the Codon Adaptation Index (CAI), which require the target to be an open reading frame, the objectives we propose are valid for every structure in the standard Eterna100 benchmark without any domain restriction.

Contribution. We propose a novel, multi-objective formulation of the RNA Inverse Folding problem. Utilizing the LEARNNA Gymnasium environment, we engineer a compound reward function with an adaptive weighting mechanism that forces the RL agent to solve the combinatorial space under strict sequence-quality constraints. The agent must find sequences that simultaneously minimize structural distance, maintain target GC-content, avoid homopolymer runs, and maximize thermodynamic stability. We provide a rigorous algorithmic comparison between value-based (DQN with intermediate reward shaping) and policy-gradient (PPO) paradigms [Roijers et al., 2013].

2 Problem Definition

2.1 The Combinatorial Optimization Problem

- **Input:** A target RNA secondary structure of length n , represented in standard dot-bracket notation (e.g., $(((((\dots))))))$).
- **Search Space:** The set of all possible nucleotide sequences of length n over the alphabet $\{A, C, G, U\}$. Size: 4^n .
- **Objective:** Find the optimal sequence S^* that maximizes a multi-objective score encompassing structural proximity, GC-content fitness, homopolymer avoidance, and thermodynamic stability.

2.2 MDP Formulation

We cast the sequence design as a finite-horizon Markov Decision Process (MDP):

- **State Space ($\mathcal{S}$):** The partially constructed sequence up to step t , concatenated with the representation of the global target structure.
- **Action Space ($\mathcal{A}$):** The discrete selection of the next nucleotide: $\{A, C, G, U\}$. $|\mathcal{A}| = 4$.
- **Transition (P):** Deterministic state update appending the chosen nucleotide.
- **Reward (R):** A shaped scalar reward with both intermediate and terminal components (Section 3.3).

3 Methods

3.1 Data Pipeline and Environment

Environment. We utilize the open-source LEARNNA Gymnasium environment [Runge et al., 2019], which wraps the C-based ViennaRNA library [Lorenz et al., 2011] for millisecond-scale thermodynamic folding evaluations, making local CPU training feasible.

Dataset. We use the full Eterna100 benchmark [Anderson-Lee et al., 2016], 100 diverse, curated RNA secondary structures of varying difficulty. Because our biological objectives (GC-content, homopolymer avoidance, MFE stability) are applicable to all RNA types, we do not need to filter or restrict the dataset.

3.2 RL Agents

Given the discrete action space and highly combinatorial nature of the problem, we evaluate two fundamentally different algorithmic paradigms:

1. **DQN (Deep Q-Network):** A value-based, off-policy method. DQN utilizes experience replay to propagate sparse, terminal rewards backwards through the sequence. To address the credit assignment difficulty inherent in long sequences with purely terminal rewards, we augment DQN with **intermediate reward shaping**: at each timestep t , the agent receives a small potential-based shaping reward proportional to the local structural compatibility of the most recently placed nucleotide (e.g., whether a valid base pair is formed where the target expects one). This follows the potential-based shaping guarantee of Ng et al. [1999], providing denser learning signals without altering the optimal policy.
2. **PPO (Proximal Policy Optimization):** A policy-gradient, on-policy method. While potentially less sample-efficient, PPO’s clipped surrogate objective often yields more stable convergence in constrained or multi-objective settings [Roijers et al., 2013]. Its on-policy nature avoids the distributional shift that may arise in DQN when reward component weights change during training.

3.3 Multi-Objective Reward Design

The terminal reward is defined as:

$$R = \alpha \cdot R_{\text{struct}} + \beta \cdot R_{\text{GC}} - \gamma \cdot P_{\text{homo}} + \delta \cdot R_{\text{MFE}} \quad (1)$$

where each component is defined as follows:

- R_{struct} (**Structural Accuracy**): Derived from the normalized Hamming distance between the target dot-bracket structure and the MFE fold of the agent’s sequence, computed by ViennaRNA [Lorenz et al., 2011]. A distance of 0 yields a reward of 1.0.
- R_{GC} (**GC-Content Fitness**): Measures how close the sequence’s GC ratio is to the biologically preferred range of 40–60%. We define:

$$R_{\text{GC}} = 1 - \frac{\max(0, |\text{GC}\% - 50\%| - 10\%)}{40\%} \quad (2)$$

yielding full reward within $[0.4, 0.6]$ and linearly decreasing penalty outside. Extreme GC-content is associated with RNA misfolding, aggregation, and reduced solubility [Leppek et al., 2022].

- P_{homo} (**Homopolymer Penalty**): Penalizes contiguous runs of the same nucleotide exceeding a threshold length k (default $k = 4$):

$$P_{\text{homo}} = \frac{1}{n} \sum_{r \in \text{runs}} \max(0, \text{len}(r) - k) \quad (3)$$

Long homopolymer stretches impair chemical synthesis fidelity, cause sequencing errors, and reduce structural diversity.

- R_{MFE} (**Thermodynamic Stability**): The normalized minimum free energy of the predicted fold. A more negative MFE indicates stronger stability:

$$R_{\text{MFE}} = \frac{|\text{MFE}|}{n} \quad (4)$$

This ensures the agent finds sequences that fold into the target with high thermodynamic confidence.

Adaptive Weight Scheduling. Static weight scalarization in multi-objective RL often leads to objective dominance, where the agent collapses onto a single objective. We address this with a phased curriculum strategy:

- **Phase A (Structural Warm-up, first 30% of episodes):** $\alpha = 1.0$, $\beta = \gamma = \delta = 0$. The agent first learns structurally valid folding, establishing a strong structural prior.
- **Phase B (Gradual Integration, next 40%):** β , γ , and δ are linearly annealed from 0 to their target values while α is slightly reduced. This prevents catastrophic forgetting of the structural objective.
- **Phase C (Joint Optimization, final 30%):** All weights are fixed. We evaluate multiple configurations via grid search over $(\alpha, \beta, \gamma, \delta) \in \{(0.5, 0.2, 0.1, 0.2), (0.6, 0.15, 0.1, 0.15), (0.4, 0.2, 0.15, 0.25)\}$ to identify Pareto-optimal trade-offs.

4 Experimental Design and Evaluation

4.1 Evaluation Metrics

- **Structural Success Rate:** Percentage of generated sequences that perfectly fold into the target structure (Hamming distance = 0).
- **Sequence Quality Score:** A composite metric over structurally successful sequences, reporting mean GC-content (and its deviation from the 40–60% band), average longest homopolymer run, and normalized MFE.
- **Sample Efficiency:** Number of episodes required for policy convergence.
- **Per-Objective Reward Curves:** Each reward component tracked independently across training to diagnose objective dominance or collapse.

4.2 Experimental Protocol

- **Phase 1 (Baseline Replication):** Train DQN and PPO on the standard single-objective LEARN reward (structure only) across 20 Eterna100 targets. Validate reproduction of published results.
- **Phase 2 (Multi-Objective Integration):** Train agents on the same 20 targets using the adaptive reward schedule with all four objectives.
- **Phase 3 (Ablation and Comparison):** (a) Ablate each reward component to quantify its individual contribution. (b) Compare DQN (with intermediate shaping) vs. PPO on convergence speed, structural accuracy, and sequence quality. (c) Evaluate adaptive vs. static weighting.

5 Timeline

The project spans 6 weeks. Weeks 3–4 are merged into a single block for reward engineering and multi-objective training, as this phase carries the highest risk of iterative debugging.

Week	Planned Tasks
1	Setup & Foundation: Acquire Eterna100 dataset, configure the local LEARN environment (resolve dependency issues if needed, or port to current Gymnasium API). Implement GC-content, homopolymer penalty, and MFE stability metric functions. Set up Stable-Baselines3 and connect DQN/PPO agents. Implement intermediate reward shaping for DQN.
2	Baseline Replication: Execute Phase 1 baseline tests. Validate reproduction of published LEARN results. Begin integrating biological metrics into a custom Gymnasium wrapper. Initial hyperparameter sweep for baseline agents.
3–4	Reward Engineering & Multi-Objective Training: Debug reward function dynamics. Implement the adaptive weight scheduling (Phases A/B/C). Execute Phase 2 multi-objective training with grid search over weight configurations. Monitor per-objective reward curves to detect objective dominance. <i>Two weeks are allocated to accommodate iterative debugging and tuning.</i>
5	Ablation & Analysis: Conduct Phase 3 ablation and trade-off analysis. Compile comparative metrics (sequence quality vs. baseline, structural accuracy trade-off). Extract sample efficiency data. Perform statistical significance tests. Compare adaptive vs. static weighting results.
6	Drafting & Finalization: Draft the final manuscript in ICLR format. Synthesize all sections. Code cleanup and documentation with explanatory comments (per course requirement).

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